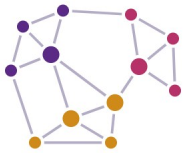
Vinh Pham

● LLM Interpretability ● Time-series/Spatio-temporal Analysis

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I am currently an undergraduate in Mathematics (Probability and Statistics) at Georgia Tech, advised by **Professor Shihao Yang (Georgia Tech)** since my freshman year. I study how sequence models represent and route temporal and relational information, from uncovering the internal mechanisms of language models to designing attention and analyzing time-series and spatio-temporal models. As an undergraduate, I have led or contributed to paper submissions to top ML Conferences as first author and co-author.

Education

Georgia Institute of Technology, Atlanta, GA Aug 2024 – May 2028

BSc in Mathematics, Probability and Statistics GPA 4.0/4.0

Relevant core coursework

Probability	Stochastic Process, Statistical Theory, Probability Theory
Analysis	Real/Complex Analysis, Advanced Linear Algebra
Computation	Convex Optimization, Computational Method for Simulation and Machine Learning
Theory	Design and Analysis of Algorithms, ML Theoretical Analysis

Research interests

Expressive Sequence-Modeling | Mechanistic Interpretability of LLMs | Time-series/Spatio-temporal Analysis

Graph Neural Network | Multimodal Foundation Models

Publications

● LLM Interpretability ** leading author*

A Narrow Residual Bottleneck from Evidence to Answers in Language Models
Vinh Pham*, Jevon Twitty, Shihao Yang, Jiecheng Lu — ICLR 2027, under review

Poke, Don't Fit: Predicting Activation Steering with Contextual Jacobians Instead of Fitted Layer Operators
Vinh Pham*, Shihao Yang, Jiecheng Lu — In preparation

● Time-series/Spatio-temporal Forecasting

StretchTime: Adaptive Time Series Forecasting via Symplectic Attention
Yubin Kim, Viresh Pati, Jevon Twitty, **Vinh Pham**, Shihao Yang, Jiecheng Lu — ICML 2026 (South Korea)

Spatial-as-Refinement: Gated Graph Operators over Strong Temporal Predictors for Scalable Spatio-temporal Forecasting
Vinh Pham*, Jevon Twitty, Shihao Yang, Jiecheng Lu — ICLR 2027, under review

Temporal Operator Attention: Recovering Common Operators in Time-Series Analysis
Jevon Twitty*, **Vinh Pham***, Nitiwith Rotchanarak, Viresh Pati, Yubin Kim, Shihao Yang, Jiecheng Lu — Preprint

CAPS: Unifying Attention, Recurrence, and Alignment in Transformer-based Time Series Forecasting
Viresh Pati, Yubin Kim, **Vinh Pham**, Jevon Twitty, Shihao Yang, Jiecheng Lu — Preprint

Talks and posters

ICML 2026, South Korea. StretchTime: Adaptive Time Series Forecasting via Symplectic Attention

ML@GT Conference 2026. A Narrow Residual Bottleneck from Evidence to Answers in Language Models

GT ISyE Summer Research. DynaTime Attention for Long-Sequence Time Series Forecasting

Research experience

Georgia Tech Deep Learning Lab

Sep 2024 – Present

Research Assistant, advised by Professor Shihao Yang

- LLM Interpretability

Poke, Don't Fit: Predicting Activation Steering with Contextual Jacobians Instead of Fitted Layer Operators
Showed that static residual stream models fail across depth both theoretically and empirically, whereas contextual Jacobian alignment unlocks much stronger LLM behavioral steering.

A Narrow Residual Bottleneck from Evidence to Answers in Language Models
Discovered a 2–3 token residual bottleneck through which LLMs route evidence to answers: ~98% pass a two-sided causal test versus ~1% of random sets, while same-size top-importance heads fail it, across frontier model families.

- Time-series/Spatio-temporal Analysis

Spatial-as-Refinement: Gated Graph Operators over Strong Temporal Predictors for Scalable Spatio-temporal Forecasting
Designed a gated graph-refinement architecture that provably eliminates multiplicative noise compounding in spatio-temporal forecasting models, achieving state-of-the-art on 14 real-world benchmarks.

StretchTime: Adaptive Time Series Forecasting via Symplectic Attention [ICML 2026]
Proposed Symplectic Positional Embeddings, a Hamiltonian generalization of RoPE with an input-adaptive time-warp module, so Transformers learn non-affine temporal stretching end-to-end, achieving state-of-the-art against newest models.

Temporal Operator Attention: Recovering Common Operators in Time-Series Analysis
Proved softmax attention is a low-pass averaging operator unable to realize signed filters, demixing projections, or phase-adaptive operators; introduced TOA, a residual, input-adaptive operator.

CAPS: Unifying Attention, Recurrence, and Alignment in Transformer-based Time Series Forecasting
A linear-complexity structured attention layer that separates global trends, local shocks, and seasonal phase alignment via SO(2) RoPE rotations and Riemann-softmax, prefix-product, and Clock gating paths.

Yale Graph and Geometric Learning Lab

May 2026 – Present

Research Assistant, advised by Professor Rex Ying

- Time-series/Spatio-temporal Analysis

Forecasting stress-response and substance-use risk from clinical monitoring data with heavy missingness, extending channel clustering with a relational GNN (with the Yale School of Medicine).

LLM-augmented trajectory forecasting on 148K taxi GPS traces: gated fusion of language-model context embeddings and RL post-training on self-generated rollouts, cutting 10-step MAE ~30%.

NASA Ames Research Center

May 2026 – Aug 2026

Deep Learning Researcher, Wildfire, Ecosystem Resilience, and Risk Assessment Initiative (WERK)

Preprocessed 12,600 raw hyperspectral satellite images and trained spatio-temporal benchmark models to classify California land cover (landmasses, water bodies), with baseline validation and tuning.

Teaching

Teaching Assistant, Georgia Tech

Math 3406, Advanced Linear Algebra — Fall 2025, Spring 2026, Fall 2026

Math 4740, Computational Method for Simulation and Machine Learning — Fall 2026

Prepare lecture notes and worked examples on vector spaces, eigenvalue theory, and matrix decompositions; grade homework and exams; hold office hours on proofs and problem solving.